Hotel & Tourism Performance Analytics Dashboard

Designed and developed a tourism analytics dashboard using Power BI to analyze visitor trends, hotel occupancy, and revenue performance in Singapore.

Project Summary

- **Project Title**
  - Hotel & Tourism Performance Analytics Dashboard
- **Objective**
  - Analyze tourism trends and hotel performance in Singapore to understand how visitor arrivals influence occupancy rates, pricing strategies, and hotel revenue.
- **Tools Used**
  - Microsoft Power BI – dashboard design and visualization
  - Power Query – data cleaning and transformation
- **Dataset**
  - Singapore tourism dataset from data.gov.sg
- **Key Outcomes**
  - Developed an interactive Power BI dashboard to analyze tourism demand trends
  - Identified the relationship between visitor arrivals, occupancy rate, and hotel revenue
  - Provided strategic recommendations for hotel performance

Context

- Tourism is a major driver of economic growth and employment.
- International arrivals fluctuate due to seasonality, global events, and market trends.
- Hotels depend on aligning capacity and pricing strategies with visitor demand.
- Stakeholders need clear insights to make data-driven decisions.

Problem Statement

- Hotels need to balance **room pricing, occupancy, and demand fluctuations** driven by tourism trends.
- However, limited visibility into tourism patterns makes it difficult for stakeholders to:
  - anticipate demand changes
  - optimize pricing strategies
  - plan hotel capacity effectively
- This project analyzes tourism data to uncover **key demand patterns and their impact on hotel performance.**

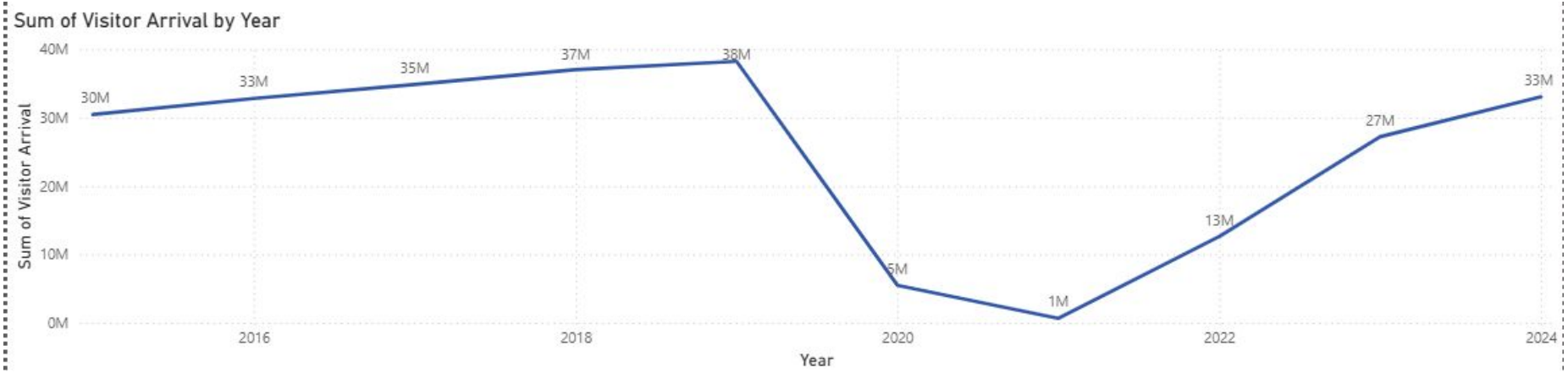
Tools & Technologies

- **Power BI** – Data visualization and dashboard development
- **Power Query** – Data cleaning and transformation
- **Data Source:** Singapore tourism statistics (data.gov.sg)
- **Techniques Used:**
  - Data transformation
  - Time-series trend analysis
  - Dashboard visualization
  - KPI analysis (ADR, occupancy rate, visitor arrivals)

Data Preparation

- Imported tourism dataset from data.gov.sg
- Cleaned and standardized the dataset using **Power Query**, including handling missing values, formatting date fields, and creating calculated metrics.
- Time period analysed: 2015–2024
- Created calculated metrics:
  - Average Daily Rate (ADR)
  - Occupancy Rate
  - YoY Growth
  - Revenue Per Available Room (RevPAR)
- Built an interactive dashboard in **Power BI**

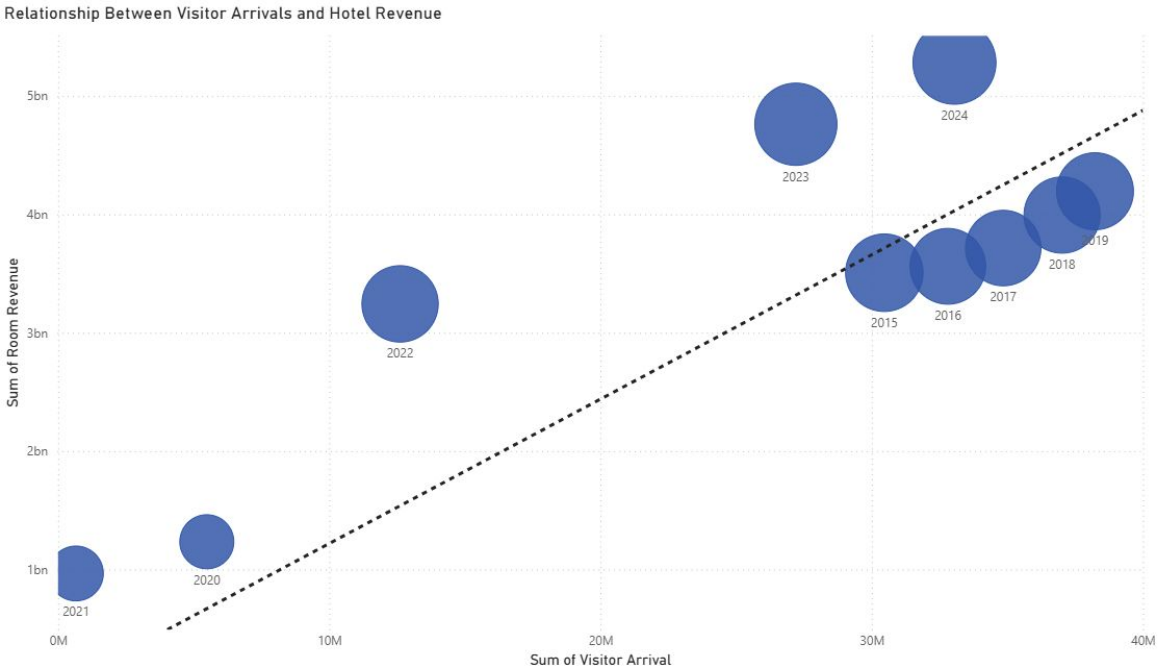
Yearly Arrival Statistics



Insights:

- *From 2015 to 2019, visitor arrivals grew steadily from 30 million to 38 million, showing strong tourism momentum.*
- *In 2020 and 2021, arrivals collapsed to 5 million and then 1 million due to COVID-19 travel restrictions and lockdowns.*
- *By 2022–2024, tourism rebounded quickly, reaching 33 million arrivals, nearly back to pre-pandemic levels.*

Correlation between Visitor Arrivals and Hotel Revenue



Insights:

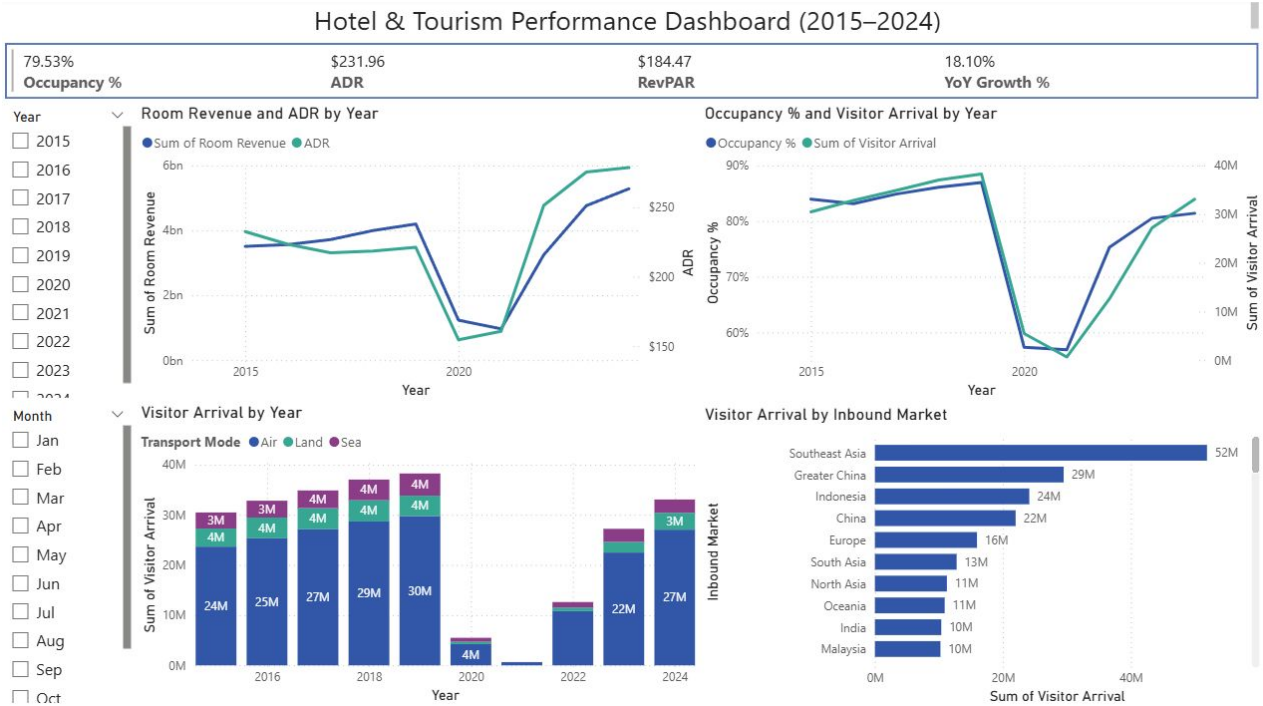
- *The scatter plot shows a **strong positive** relationship between visitor arrivals and hotel revenue.*
- *As tourism demand **increases**, hotels experience **higher** room revenue, highlighting the importance of tourism flows in driving hotel performance.*

Dashboard Overview

- provides an interactive view of tourism and hotel performance, allowing stakeholders to monitor:

Occupancy Rate:
Track hotel capacity utilization over time

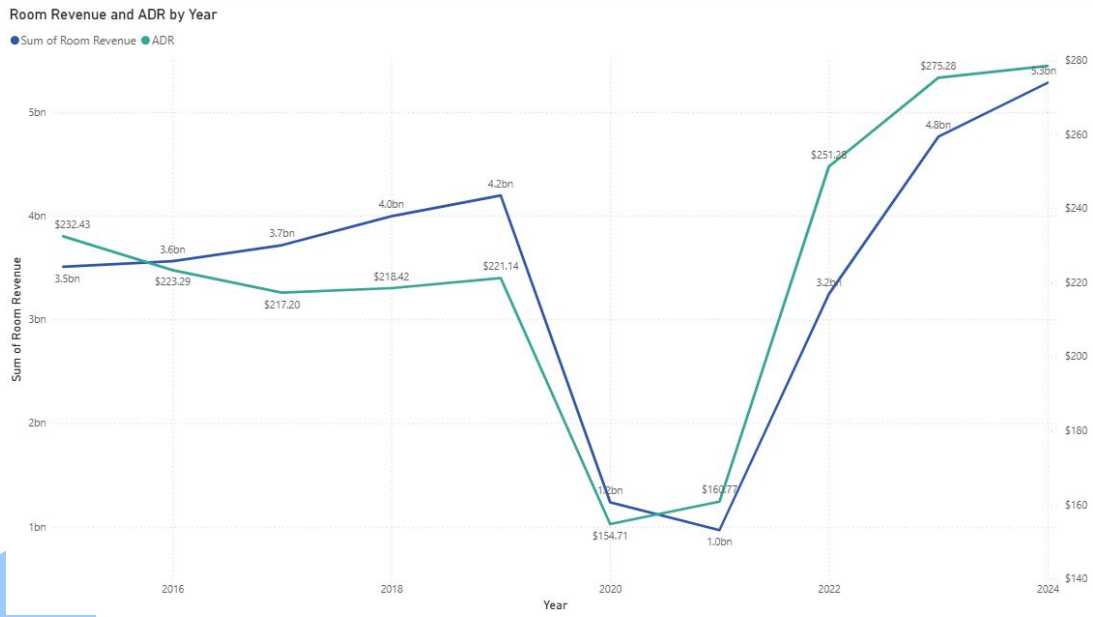
Average Daily Rate (ADR):
Track hotel pricing strategy trend



Revenue Per Available Room (RevPAR):
Measures how effectively hotels generate revenue from available rooms.

YoY Growth:
Shows long-term performance trend

How does ADR (Average Daily Rate) affect total room revenue?



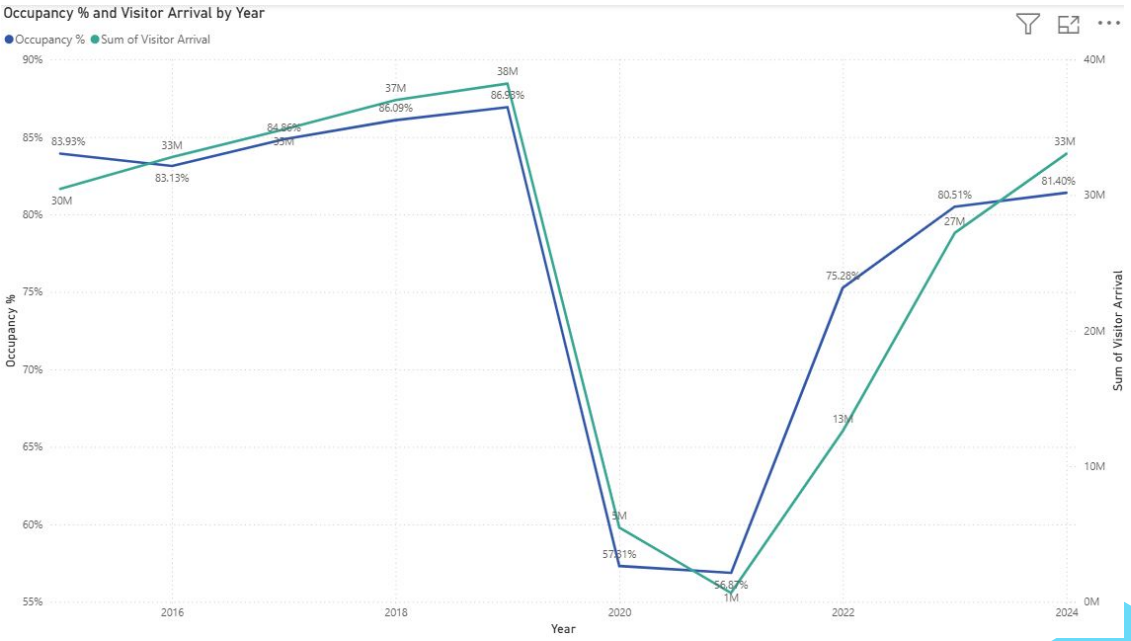
Insights:

- *Room revenue rose from about **$3.2B to $4.8B**, while ADR increased from roughly **$180 to $240**, showing strong demand and pricing power.*
- *Revenue dropped sharply to around **$1.2B**, and ADR fell to about **$150**, reflecting the severe hit from travel restrictions*
- *By 2023 ADR climbed to nearly **$280**, and revenue approached **$5B**, signaling regained demand and stronger pricing strategies.*

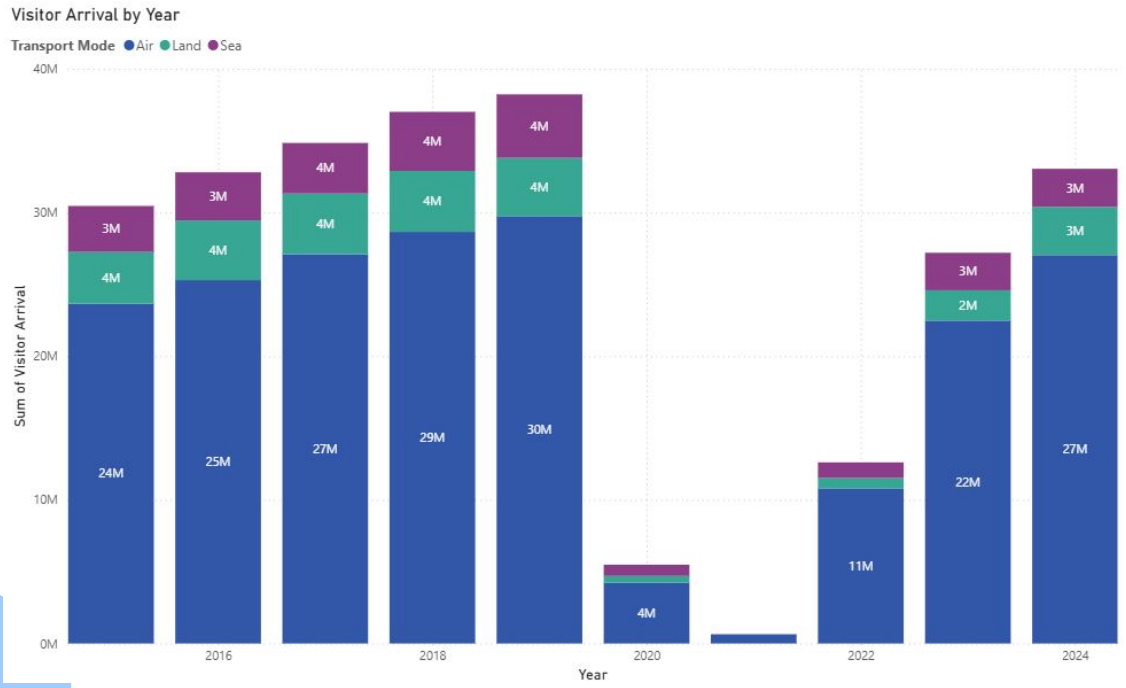
How do shifts in visitor arrivals influence occupancy rates?

Insights:

- *Occupancy remained high, rising from **83.9% in 2015** to a peak of **86.9% in 2019**, showing strong demand and efficient capacity use*
- *Occupancy dropped sharply to **51.3%**, reflecting global travel restrictions and reduced hotel stays*
- *Occupancy rebounded to **75.3% in 2022**, **80.5% in 2023**, and **81.4% in 2024**, approaching the stability seen before the pandemic*



Which transport mode contributes the most to tourism arrivals?



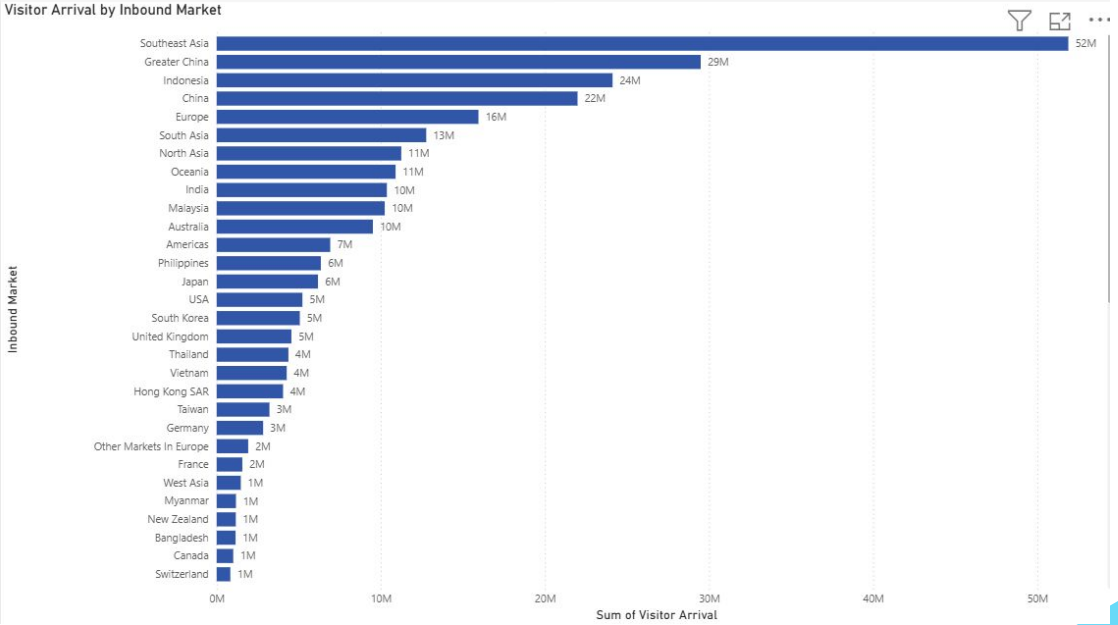
Insights:

- *Visitor arrivals collapsed across all transport modes during COVID-19, but recovery has been* ***driven primarily*** *by* ***air travel****. This indicates Singapore's tourism recovery depends heavily on international flight connectivity.*
- *By 2024, air arrivals reached 27M, while land and sea stabilized at 2–3M, highlighting the dominance of air transport in tourism recovery.*

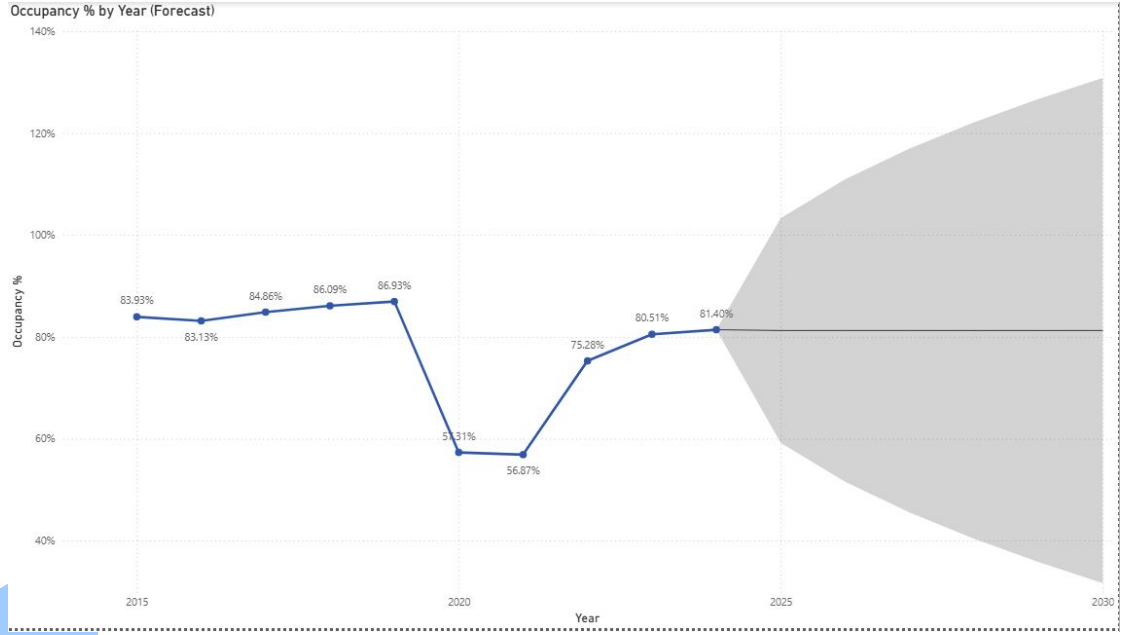
Which inbound markets drive the highest visitor volumes?

Insights:

- *The majority of arrivals come from Asia, especially **Southeast Asia** of 52M, highlighting geographic proximity and cultural ties as key drivers.*
- *Heavy reliance on Southeast Asia means **vulnerability to regional shocks.***
- *Expanding engagement with Europe, Oceania, and the Americas could stabilize long-term growth.*



Forecasting model for Occupancy until 2030



Insights:

- *The central forecast line stays at ~81%, suggesting a return to pre-crisis stability. The shaded confidence band widens over time, suggest that* ***uncertainty increases the further out it project****.*
- *By 2030,, the range spans roughly 30%–130%, showing possible upside or downside scenarios.*

Key Insights Summary

- Tourism demand strongly drives hotel performance, with occupancy consistently above **80% during stable periods**
- **ADR increased significantly after the pandemic**, indicating hotels shifted towards higher pricing strategies during recovery
- Growth momentum: YoY revenue growth >15%.
- **Air travel** dominates tourism arrivals, accounting for the majority of inbound visitors to Singapore.
- **Tourism demand is concentrated in Southeast Asia**, creating exposure to regional economic shocks.
- Diversifying visitor markets could **stabilize long-term tourism growth and hotel revenue.**

Impact of Tourism on Hotel Performance

- **Occupancy**: Tourism flows directly drive hotel occupancy. Visitor arrivals rebounding after 2021 lifted occupancy back above **80% by 2023–2024**, restoring stability.
- **Room Revenue**: More tourists mean more rooms sold. Combined with higher ADR, revenue surged to nearly **$5.3B by 2024**.
- **ADR (Average Daily Rate)**: Strong tourism demand allowed hotels to raise ADR to **$280**, showing regained pricing power..
- The 2020 collapse (occupancy down to **57%**) highlights hotels' dependence on tourism and exposure to external shocks.

Recommendations on Hotel Performance

- Maintain occupancy above **80%** while gradually increasing ADR to sustain revenue growth.
- Reduce reliance on Southeast Asian markets by expanding tourism campaigns in **Europe, Oceania, and the Americas**.
- Develop **crisis-management plans** and **flexible pricing strategies** to withstand shocks like pandemics.
- Create **loyalty programs, premium services, and bundled packages** to convert tourism demand into repeat stays and higher per-visitor spend.
- **Collaborate** with airlines and tourism boards to secure steady visitor flows and maximize hotel performance

Limitations of analysis

- Analysis relies on aggregated tourism data rather than individual hotel data.
- External factors such as economic conditions and policy changes may influence tourism trends.
- Monthly seasonality patterns may require deeper analysis using more granular datasets.

Future Improvement

- Integrate **real-time travel data** from airlines and booking platforms.
- Expand analysis with **machine learning models** to predict occupancy rates.
- Include **hotel segmentation analysis** (luxury vs budget hotels).

Conclusions

- Tourism is the **key driver** of hotel performance
- Visitor arrivals fuel **occupancy**, while ADR sets **pricing power**. Together, they determine **room revenue growth**.
- COVID-19 showed hotels' **vulnerability**, but recovery proved their **resilience**.
- Balancing **volume (occupancy)** with **value (ADR & revenue per visitor)** is essential for sustainable success.

Reference

- Singapore Government. (2025). Hotel & Tourism statistics dataset. https://data.gov.sg
- The Role of Tourism in Hospitality Management https://www.bolc.co.uk/bolc-blog/tourism-in-hospitality-management
- McKinsey & Company (2024). The state of tourism and hospitality 2024 https://www.mckinsey.com/industries/travel/our-insights/the-state-of-tourism-and-hospitality-2024